

Buttock pain

with ADAM MEAKINS

Running has substantially shaped human evolution, it is one of the most transforming events in human history. We have literally evolved to run and our big incredible butts are evidence of this.

Professor Dan Lieberman, evolutionary biologist, Harvard University

The assessment and diagnosis of an athlete with buttock pain can be a challenge for the clinician due to the complex anatomy as well as the many different structures that can refer pain into this area. Buttock pain can be experienced by a wide variety of athletes, but higher incidence is often seen in those who are involved in running, sprinting and kicking sports. Buttock pain can be either sudden or gradual in onset, it can occur in isolation or with low back, posterior thigh and even groin pain, and it can also be due to direct trauma from falls or collisions. A list of causes of buttock pain can be found in Table 30.1. Anatomy of the buttock region can be found in Figure 30.1.

CLINICAL APPROACH

Assessment of any reported buttock pain should always first try to ascertain if there is any involvement from the lumbar spine. This can be challenging due to the complex nature of low back pain especially in chronic states. For

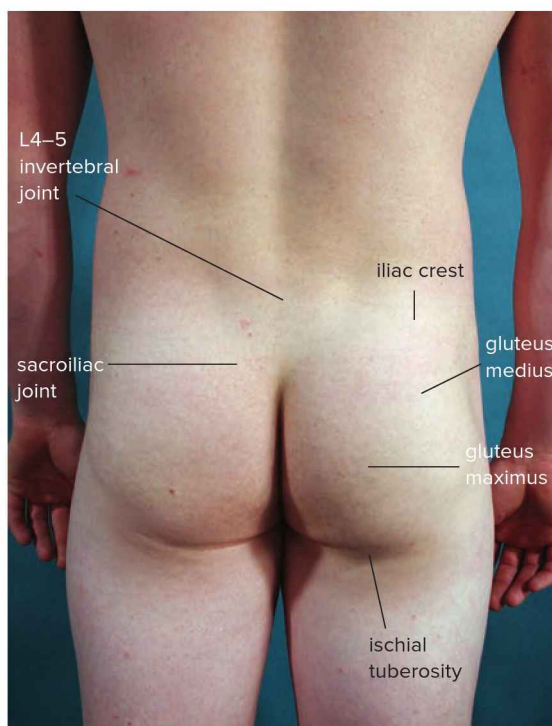
more on low back pain and lumbar spine assessment and treatment see Chapter 29.

Detailed, comprehensive and careful questioning around the onset, duration and nature of the buttock pain, together with identifying the aggravating and easing factors will help guide the clinician to probable causes and help plan the physical examination.

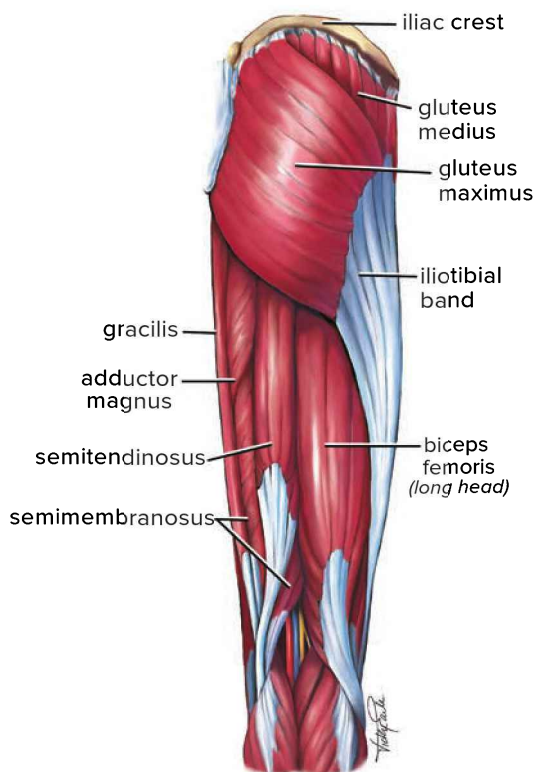
However when taking a history it is important for the clinician to focus not only on the painful area, but to recognise the many other factors that can contribute to injury and pain in athletes. Time should be taken to explore and obtain a detailed history of an athlete's lifestyle and other psychosocial factors. For example, lack of sleep, chronic levels of stress and fatigue have been found to contribute to injuries and recovery in athletes.¹⁻⁴ It is also important for a clinician to assess an athlete's psychological status. High levels of fear of re-injury, anxiety and catastrophisation have been shown to affect assessment and treatment outcomes.^{5, 6}

Table 30.1 A list of causes of buttock pain

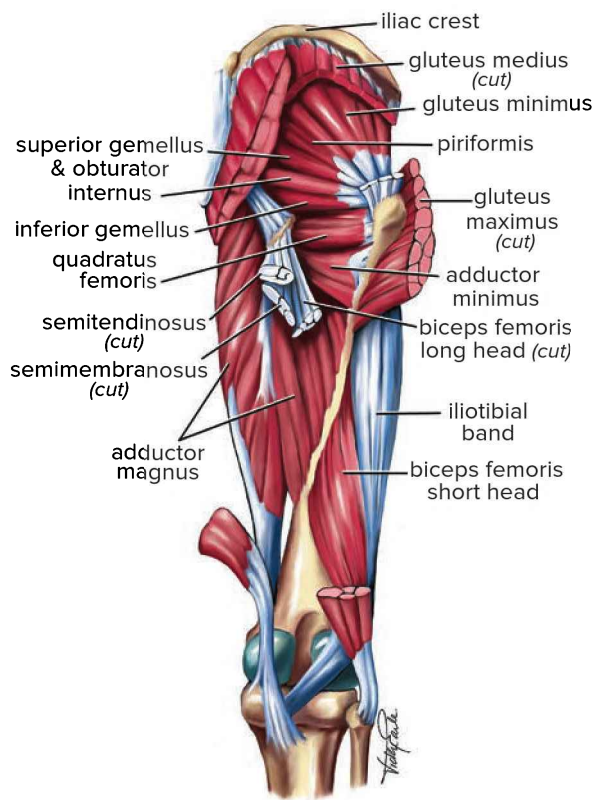
Common	Less common	Not to be missed
Myofascial pain	Piriformis syndrome	Spondyloarthropathies
Referred from lumbar spine	Ischiofemoral impingement	Ankylosing spondylitis
Proximal hamstring tendinopathy	Posterior thigh compartment syndrome	Psoriatic arthritis
Sacroiliac joint dysfunction	Ischial tuberosity injury	Reactive arthritis (Reiters)
	Proximal hamstring tendon rupture	Stress fracture sacrum
	Avulsion fracture (children)	Stress fracture pubic ramus
	Gluteus medius tendinopathy (see Chapter 31)	Malignancy
		Bone/joint infections



(a)



(b)



(c)

Figure 30.1 Anatomy of the buttocks (a) Surface anatomy (b) Superficial muscles (c) Deep muscles

There are many physical examination tests described in the literature to diagnose specific structures or pathologies of buttock pain. However, the clinician should exercise caution when a positive or negative test is found in isolation, as the specificity and sensitivity (see below) of most of these tests are poor. A combination of objective tests combined with a good thorough subjective history will give the clinician much more confidence in his or her diagnosis.

History

The onset of buttock pain can be reported either as sudden during a sporting activity or as a gradual insidious onset afterwards. Buttock pain that is deep, diffuse and hard to localise, accompanied with low back pain may suggest some lumbar spine involvement. However, it must be remembered that buttock pain without low back pain does not mean the lumbar spine is not involved.

Buttock pain combined with posterior thigh pain, with or without low back or neural symptoms, can be suggestive of a lumbar radiculopathy. However, consideration must also be given to facet joint and hamstring pathology.

Buttock pain felt around the ischial tuberosity (Fig. 30.1) or Tuber area (Fig. 30.2) may be suggestive of proximal hamstring pathology and/or ischiogluteal bursitis. Buttock pain felt more axially or directly over the sacroiliac joint, sometimes referred to as the Fortin area (Fig. 30.2), may suggest a sacroiliac joint or ligament dysfunction.⁷ However, location of pain alone should never be used in isolation to make a diagnosis.

Buttock pain reproduced with movements and activities is also not diagnostic in isolation as many structures can be aggravated concurrently. However, it may give clues to possible structures as a source of pain. Suspicion of lumbar spine involvement is increased if low back, buttock or posterior thigh pain is either increased or decreased on repeated lumbar spine movements.^{8,9}

Diurnal variation of buttock pain can be useful to establish a diagnosis. Morning pain and stiffness that lasts more than 30 minutes and eases with light activity is highly suggestive of inflammatory conditions such as sacroiliitis and ankylosing spondylitis, although further tests are needed to confirm.

Physical examination

A visual inspection of the buttock and pelvic area with the lumbar spine and upper thigh is essential. Observation of bruising around the ischial tuberosity and down the posterior thigh may indicate a proximal hamstring avulsion in those with a history of severe sudden traumatic onset of buttock pain when sprinting or running.

Observation of muscle atrophy and gross asymmetry should be noted. However, there are questions about

the clinical usefulness of many of the commonly used observational tests to assess spinal curves and pelvic positions and postures. It is known that there are wide normal variations in skeletal morphology, posture and positions in the population.¹⁰⁻¹² Despite popular belief there is no evidence that any spinal or pelvic position or posture is more associated with pain or a risk of increased incidence of injury.¹³⁻¹⁷

Palpation of the lumbar spine, pelvis and buttocks for pain provocation may assist in the identification of a pathological structure, but again it cannot be used in isolation to make a diagnosis. Palpation tests to determine hypermobility or hypomobility of the spine and pelvic joints are commonly used, but are of questionable clinical utility. The very small movements that do occur around the sacroiliac joint make the ability to detect these with palpation highly unlikely.¹⁸

The reliability of all pelvic palpation tests used to assess sacroiliac joint movement is poor.^{19,20} Palpation tests used to detect joint stiffness or laxity have been shown to have wide variation in forces applied by therapists.²¹ When combined with wide variations in individual stiffness, it makes any interpretation of joint hyper- or hypomobility completely unreliable and down to individual clinician perception.²² Finally it has been shown there is no correlation between spinal joint stiffness and pain.²³

Neurodynamic examination tests (Chapter 15) may be included if there is a suspicion of lumbar radiculopathy as a cause of buttock pain. However, there are issues with false positives and the diagnostic accuracy of these tests, meaning again they cannot be used in isolation.²⁴

Physical examination of the buttock can be divided into observation, movement and resistance tests, both isolated and dynamic.

1. Observation static
 - a. stance
 - i. alignment (shifted, inability to stand straight)
 - ii. atrophy (glutes/hamstrings)
 - iii. bruising (ischial tuberosity/posterior thigh)
 - b. single-leg stance
 - i. pain
 - ii. balance
 - iii. Trendelenburg sign
2. Observation dynamic (checking for pain/control/alignment)
 - a. gait
 - i. antalgic/Trendelenburg
 - b. squat
 - c. single-leg squat
 - i. valgus knee
 - ii. hip internal rotation
 - iii. trunk compensation

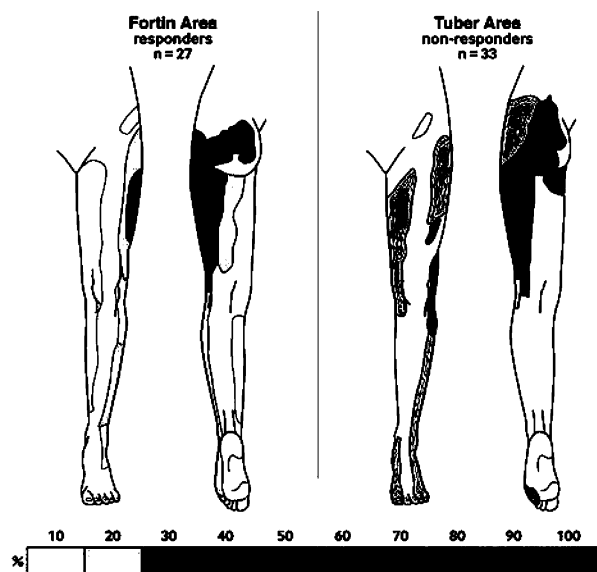


Figure 30.2 Fortin and Tuber pain areas

- d. Lunge
 - i. forward
 - ii. backward
 - iii. lateral
3. Active movements–lumbar spine
 - a. standing flexion
 - b. standing extension
 - c. standing single-leg extension (pars interarticularis test)
 - d. standing lateral flexion
 - e. standing and seated rotation
 - f. combined/functional movements
4. Active movements–hip, standing
 - a. flexion
 - b. extension
 - c. abduction
 - d. adduction
5. Active movements–hip, supine
 - a. flexion (Fig. 30.3a)
 - b. abduction
 - c. adduction
 - d. FABER
6. Passive movements–hip, supine
 - a. log roll
 - b. hip quadrant
 - c. FABER
 - d. FADIR
7. Passive movements–hip, prone
 - a. extension (Fig. 30.3b)
 - b. external rotation
 - c. internal rotation
8. Resisted movements–supine
 - a. hip flexion (psoas)
 - b. straight leg raise (rectus femoris)
 - c. external rotation (in 90° flexion)
 - d. internal rotation (in 90° flexion)
 - e. double-leg bridge (Fig. 30.3c)
 - f. single-leg bridge
9. Resisted movements–side-lying
 - a. hip abduction in neutral (TFL/glute med) (Fig. 30.3d)
 - b. hip abduction in 30° flexion (glute med)
10. Resisted movements–prone
 - a. hip extension (glute max/hamstrings)
 - b. knee flexion (hamstrings)
11. Palpation
 - a. sacroiliac joint (Fortin area)
 - b. ischial tuberosity (Tuber area) (Fig. 30.3e)
 - c. gluteal muscles
 - d. lumbar spine vertebral segments
 - e. lumbar spine paraspinal muscles



(a)



(b)



(c)

Figure 30.3 Examination (a) Active hip flexion with knee at 90° (b) Passive hip extension (c) Double-leg bridge



(d)

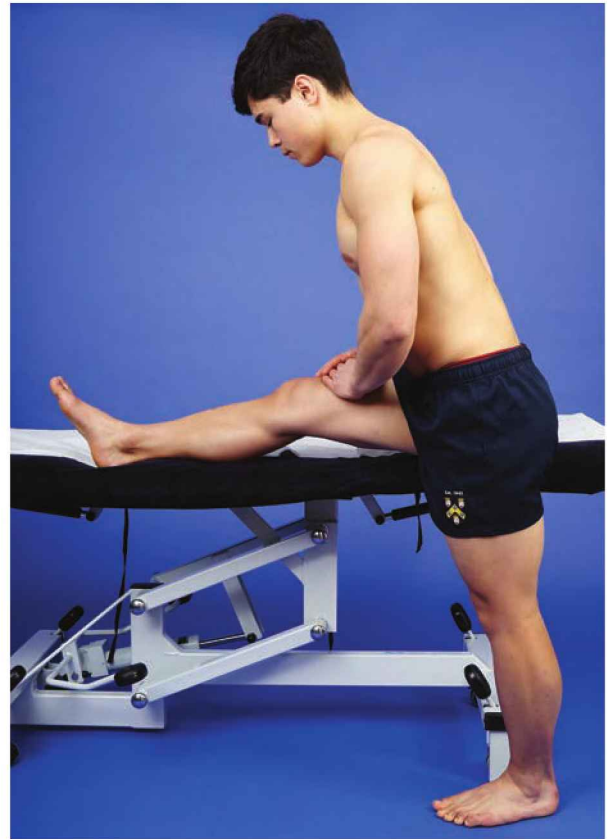


(e)



(f)

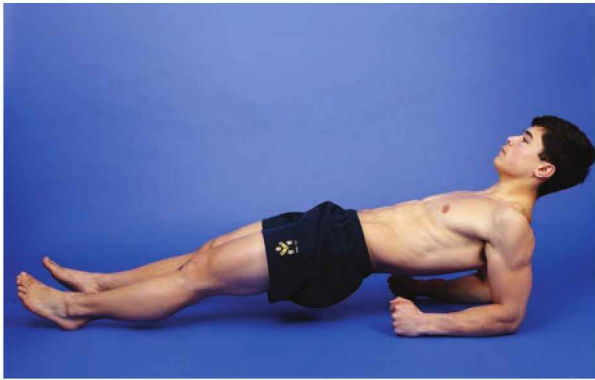
Figure 30.3 (cont.) (d) Resisted hip abduction in side-lying, hip in neutral (e) Palpation of the ischial tuberosity (f) Modified bent knee stretch



(g)

Figure 30.3 (cont.) (g) Puranen-Ovara stretch test

12. Neurodynamic tests (Chapter 15)
 - a. slump
 - b. straight leg raise
 - c. special tests—proximal hamstring tendinopathy
 - d. bent knee stretch test
 - e. modified bent knee stretch test (Fig. 30.3f)
 - f. Puranen–Ovara stretch test (Fig. 30.3g)
 - g. supine plank test (Fig. 30.3h)
13. Special tests—sacroiliac joint
 - a. Gaenslen's torque test (Fig. 30.3i)
 - b. Patrick's test (FABER) (Fig. 30.3j)
 - c. femoral shear/thigh thrust test (Fig. 30.3k)
 - d. distraction test (Fig. 30.3l)
 - e. compression test (Fig. 30.3m)
 - f. sacral thrust test (Fig. 30.3n)
 - g. active straight leg raise test (Fig. 30.3o)
14. Special tests—piriformis
 - a. active side-lying piriformis test (Fig. 30.3p)
 - b. passive seated piriformis test (Fig. 30.3q)
 - c. FAIR test (Fig. 30.3r)



(h)



(k)



(i)



(l)



(j)



(m)

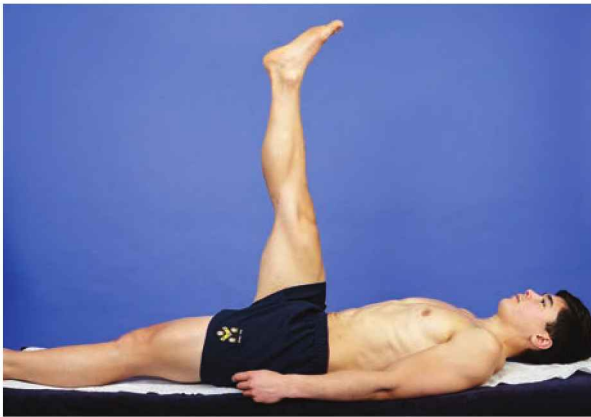
Figure 30.3 (cont.) (h) Supine plank test (i) Gaenslen's torque test (j) Patrick's test (FABER) (k) Femoral shear/thigh thrust test (l) Distraction test (m) Compression test



(n)



(q)



(o)



(r)



(p)

Figure 30.3 (cont.) (n) Sacral thrust test (o) Active straight leg test (p) Active side-lying piriformis test or Beatty test (active resisted abduction with patient side-lying with the hip flexed 90°)

Figure 30.3 (cont.) (q) Passive seated piriformis or PACE test (active resisted abduction with patient seated with hip flexed 90°) (r) FAIR (passive flexion, adduction and internal rotation of the hip, with patient in supine)

Sensitivity and specificity of tests for buttock pain

Refer to Table 30.2.

Key outcome measures

There is no validated outcome measure for general buttock pain. However, the VISA-H is a validated outcome measure that can be used for monitoring progress for those with proximal hamstring tendinopathy.²⁵ It consists of eight questions to assess severity of symptoms, function and sporting activity.

Investigations

Pelvic radiographs can be used if there is suspicion of bony trauma as the cause of buttock pain such as a pubic ramus

Table 30.2 Specificity and sensitivity of tests for buttock pain

	Sensitivity	Specificity	Positive likelihood ratio	Negative likelihood ratio
<i>Proximal hamstring tendinopathy special tests</i>				
Bent knee stretch test	0.84	0.87	6.5	0.18
Modified bent knee stretch test	0.89	0.91	10.2	0.12
Puranen-Ovara stretch test	0.76	0.82	4.2	0.29
Supine plank test	NA	NA	NA	NA
<i>Sacroiliac joint special tests</i>				
Gaenslen's torque test	0.50	0.77	2.21	0.65
Patrick's test (FABER)	0.40	0.99	NA	NA
Femoral shear/thigh thrust test	0.88	0.69	2.80	0.18
Distraction test	0.60	0.81	3.20	0.49
Compression test	0.69	0.69	2.20	0.46
Sacral thrust test	0.63	0.75	2.50	0.50
Active straight leg raise test	0.58	0.97	NA	NA
Van der Wurff cluster (3 or more +ve from tests 1, 2, 3, 4, 5)	0.85	0.79	4.02	0.19
Laslett's cluster (2 or more +ve from tests 3, 4, 5, 6)	0.88	0.78	4.00	0.16
<i>Gluteal sciatic nerve entrapment/piriformis syndrome special tests</i>				
Straight leg raise	0.15	0.95	3.20	0.90
Active side-lying piriformis test	0.78	0.80	3.90	0.27
Passive seated piriformis test	0.52	0.90	5.22	0.53
FAIR test	0.88	0.83	5.20	0.14

fracture or ischial tuberosity avulsion in the younger athlete. Loss of sacroiliac joint definition on a pelvic X-ray strongly suggests a spondyloarthropathy, but needs to be confirmed with blood tests and MRIs.

Lumbar X-rays can be used if a pars interarticularis fracture is suspected; however, these can be difficult to see and oblique views are best when looking for the 'Scotty dog' collar sign (see Chapter 29).

If there is a suspicion of a stress fracture or reaction of the pars interarticularis, ishium, scarum or pubic rami, then magnetic resonance imaging (MRI) should be considered. Isotopic bone or CT scans may also be an option but careful consideration should be given to exposure to radiation, especially in the younger athlete. Caution must also be used when interpreting imaging, because joint problems such as spondylosis or spondylolistheses can be very common in asymptomatic athletes.²⁶

MRI can be used to confirm other soft tissue structures as a cause of buttock pain such as ischiogluteal bursitis, proximal hamstring tendinopathy and prolapsed

intervertebral disc. Again, caution is advised when interpreting imaging of the lumbar spine, as many so-called pathological findings are often seen in asymptomatic patients.²⁷

Diagnostic ultrasound scanning can be used to assess tendons and bursa around the buttock area. However, they are highly user dependent and the reliability of the findings is based on the clinician's skill and interpretation.²⁸

MYOFASCIAL BUTTOCK PAIN

Strains to the lumbar, buttock and posterior thigh muscles, sacroiliac ligaments and other soft tissues are arguably the most common cause of back and buttock pain in athletes. It is estimated that up to 97% of all sporting injuries are soft tissue in nature.²⁹ A soft tissue injury is often reported as a fairly localised pain that is aggravated by continued activity or resistance.

The buttocks provide locomotion and power, but also have important stabilising roles. The buttock muscles can often become overloaded, strained and painful during

sports, and this may be due to poor training program design, training error, or a biomechanical deficit or muscle weakness. Palpation of the soft tissue is used to locate areas of myofascial pain, feeling for localised areas of tightness, muscle knots or taut bands. These are commonly referred to as 'trigger points' (see Chapter 4) and may be adverse areas of sustained muscle fibre contractions.³⁰

Examination

A lack of gluteal muscle strength can be a common cause for buttock pain in an athlete. A common sign of gluteal muscle weakness is the Trendelenburg sign.³¹ This is a drop of the contralateral pelvis, or a compensated hitch or lateral lean of the trunk when standing on one leg. This can be difficult to notice in fit and strong individuals, and can sometimes only be identified when the athlete is fatigued or asked to stand on one leg for extended periods.³²

To assess the strength of the gluteal muscles in athletes, isometric manual muscle testing can be used. For more accurate objective assessment and monitoring, hand-held dynamometry has been shown to have good reliability.³³ However, there are limitations on hand-held

dynamometry testing due to individual examiner strength differences.³⁴

There is debate around the best positions to assess and strengthen the gluteal muscles.^{35, 36} However, resisted side-lying hip abduction (Fig. 30.3d) is seen to produce high levels of activation in the gluteus medius and resisted prone hip extension produces high levels in the gluteus maximus.³⁷ These clinical tests are simple, quick and easy to perform.

Treatment of myofascial buttock pain

If weakness is found in the gluteal muscles, a well-rounded, progressive strength and conditioning program needs to be adopted to address it that incorporates both isolated and functional multi-joint compound exercises in weight-bearing positions. For more on exercise prescription see Chapter 10. The levels of EMG activity of various exercises involving the gluteal muscles are shown in Figure 30.4.³⁸

Treatment for tender tight muscle bands or 'trigger points' is often performed with soft tissue techniques such as ischaemic pressure with thumb or elbow, or via invasive techniques such as dry needling or injections.³⁹

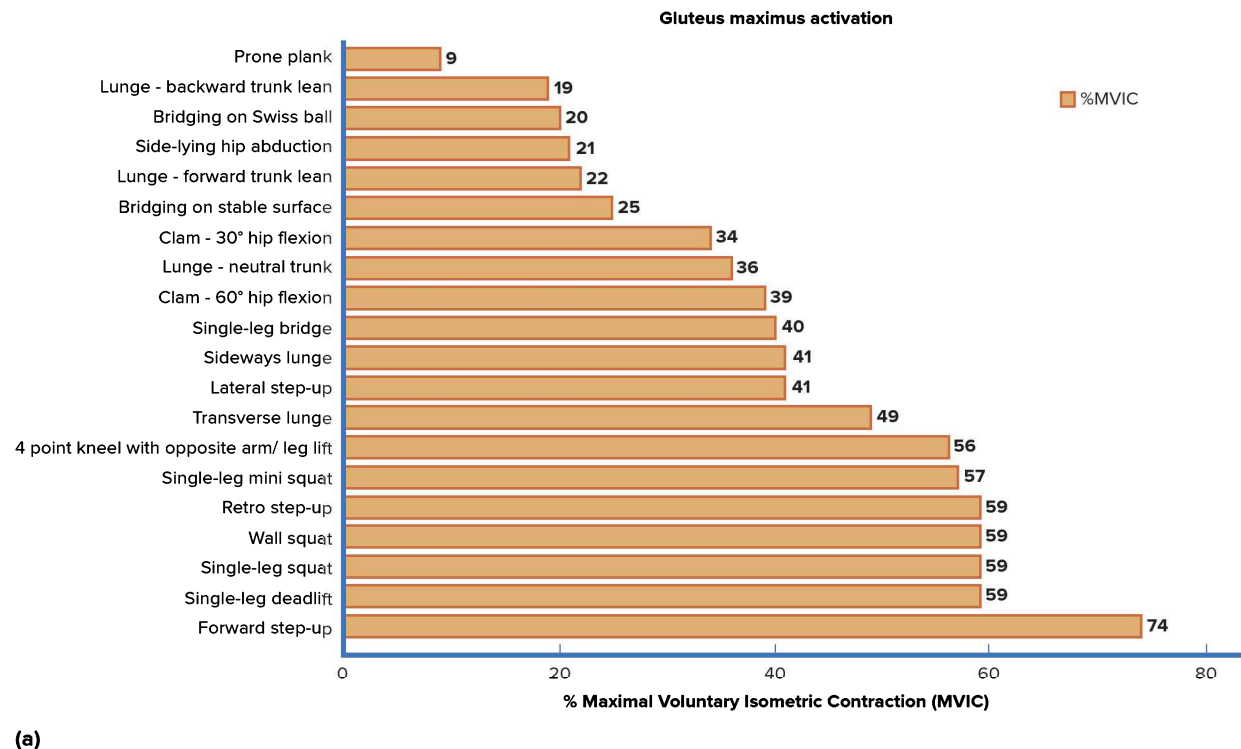
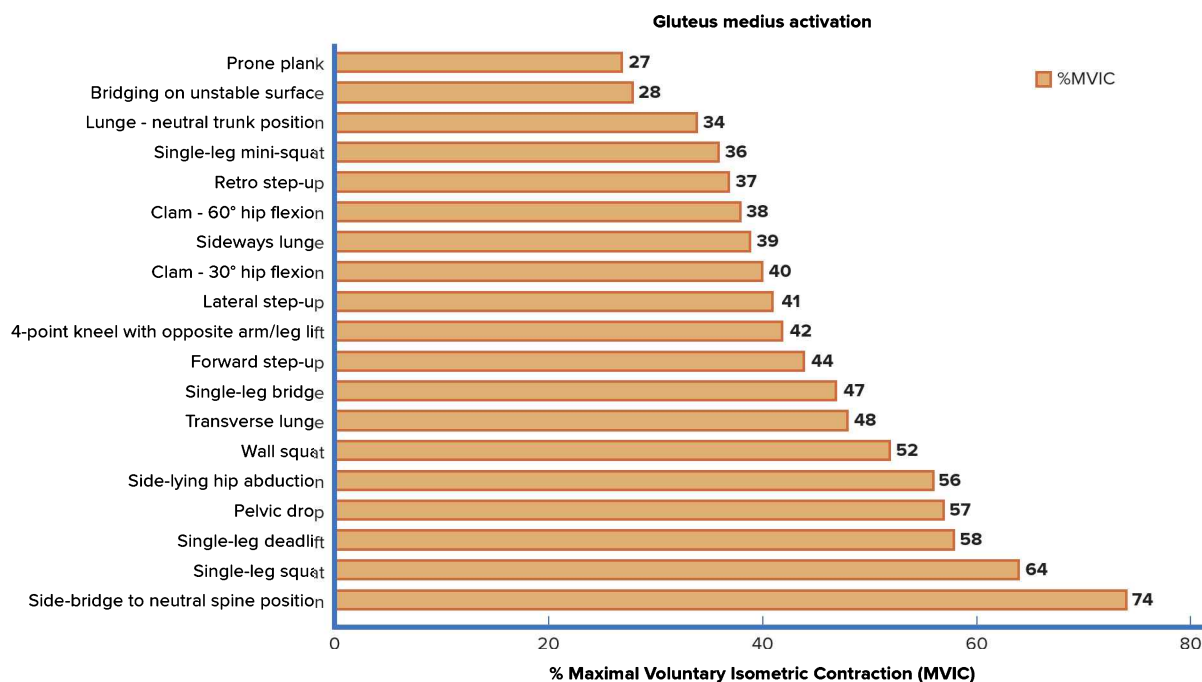


Figure 30.4 Exercises and the levels of electromyographic activity of (a) Gluteus maximus



(b)

Figure 30.4 (cont.) Exercises and the levels of electromyographic activity of (b) Gluteus medius³⁸

REFERRED PAIN FROM THE LUMBAR SPINE

Buttock pain can often be referred from the lumbar spine and be associated with or without concurrent low back pain. For more on the assessment and treatment of the lumbar spine see Chapter 29.

The most common lumbar spine structures that can cause buttock pain are the zygapophyseal (facet) joints or intervertebral discs. An athlete with either of these may describe a dull, deep diffuse pain in the buttock that can vary in intensity and location.

Pain may be aggravated with repeated lumbar spine movements as well as prolonged static postures, such as sitting or standing. Palpation of the vertebral segments and paraspinal muscles may provoke local and referred pain.

Examination

Repeated movements of the lumbar spine in one direction can cause low back and buttock pain to centralise, and when repeated in the other direction can cause the pain to peripheralise. This is highly suggestive of a discogenic lumbar radiculopathy.⁹ Lumbar facet joint pathology can be almost conclusively ruled out if seated lumbar extension and combined rotation testing is negative.^{40, 41}

Lumbar radiculopathy can also present with neural symptoms, such as pain and paraesthesia into the posterior

thigh or further distally. Buttock pain caused by nerve root compression or irritation can be aggravated with neurodynamic testing, such as the slump and straight leg raise tests (Chapter 15).⁴²

However, it has been shown that pain that is not specific to lumbar radiculopathy, such as that associated with hamstring issues, may lead to false positives for the straight leg raise test and can inflate the sensitivity of these tests.²⁴

Treatment

Treatment of lumbar spine disorders requires an integrated multimodal approach to reflect the complex nature of low back pain. This can include passive treatment, such as manual therapy, but the primary focus should always be on active rehabilitation looking to restore mobility, movement and strength, with consideration given to any psychological factors, such as fear or apprehension that can contribute to increased central sensitisation. An individualised progressive strength and conditioning program should incorporate clear patient-centred goals and objectives based on a realistic schedule for increasing exposure, duration and intensity to activity, with the ultimate goal being to return the athlete to play and competition. See Chapter 29 for more on treatment of low back pain.

PROXIMAL HAMSTRING TENDINOPATHY

The hamstrings are a commonly strained and injured muscle group during sporting activity. Although not as common as acute hamstring muscle tears (Chapter 34), proximal hamstring tendinopathy is a commonly encountered problem in sports that involve long distance running, jumping or repeated hip flexion. Tendinopathy has an unclear pathogenesis and there is even a lack of consensus over terminology; however, it is usually characterised as an activity overload-related condition culminating in pain, impaired performance and extended periods of rehabilitation.⁴³

Functional anatomy

The attachment of the three hamstrings onto the pelvis and the ischiogluteal bursa is complex (Fig. 30.5). All three hamstring tendons blend into a common tendon before they insert onto the ischial tuberosity. The hamstrings cross both the hip and knee joint, which means they are exposed to large eccentric tensile forces during activities that involve both hip flexion and knee extension, such as when decelerating the leg and foot to place it on the ground in the late swing phase of running.

The proximal hamstring tendon is also subject to high compressive loads when the hip is flexed (Fig. 30.6). This compression of the proximal hamstring tendon against the ischial tuberosity is absorbed and attenuated by the ischiogluteal bursa that is positioned between them.

The actions of the hamstrings are classically described as knee flexion and hip extension. However, as they also attach onto the tibia, their role in lower limb function and biomechanics, including the foot and ankle, cannot be overlooked.⁴⁴

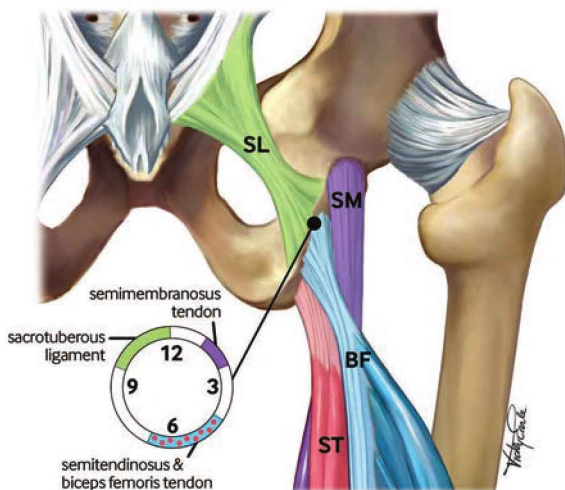


Figure 30.5 Proximal hamstring anatomy

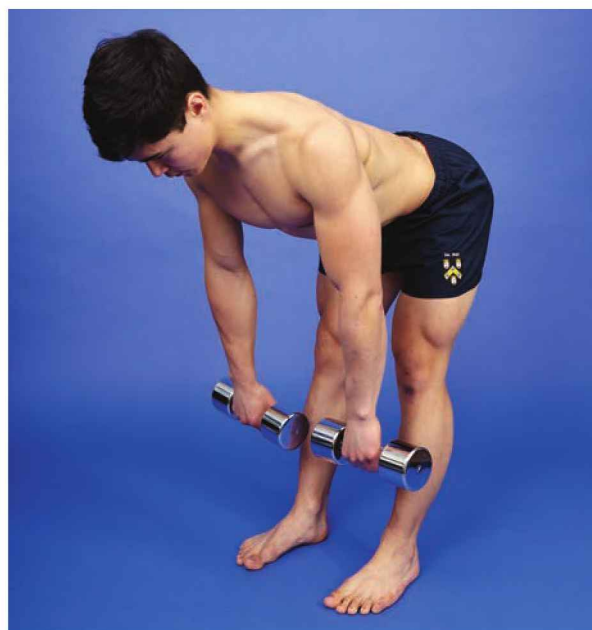


Figure 30.6 High compressive loads of proximal hamstring with the hip in flexion. Ensure spine is straight

Proximal hamstring tendinopathy (PHT) was first described in 1988 as 'high hamstring syndrome',⁴⁵ with the common features of pain felt around the lower gluteal region and down the posterior thigh. PHT occurs frequently in association with ischiogluteal bursitis, but no clinical test can differentiate between these two conditions, except for MRI or ultrasound imaging. However, as the mechanisms and subsequent management are similar for both of these conditions, apart from the use of injection therapy for recalcitrant ischiogluteal bursitis, there is no need to separate them clinically.

PHT is experienced by athletes across a wide variety of sports, but it is often seen in middle and long distance runners, and occasionally sprinters.⁴⁶ Initially, PHT pain is mild during the sport or activity, but it soon progresses and completely stops the athlete from performing. PHT pain can then progress to be felt in normal daily activities, such as walking and climbing stairs, sitting for prolonged periods or when stretching the hamstrings or gluteal muscles.

PHT is often misdiagnosed in the early stages as a pulled hamstring muscle or mild sciatica. Early identification of PHT is critical, however, as it often progresses rapidly and recovery times can extend significantly if it is not managed appropriately in the early stages.

The primary cause of PHT is mechanical overload of the tendon, and the most common factor in its aetiology is a sudden change or increase in training duration, intensity or method. Therefore, a thorough and detailed subjective history of the athlete's training history is essential.

Attention should be given to any sudden increase in amount or intensity, or to activities that involve greater hip flexion movements, such as hill sprints, weighted deadlifts, lunges or squats. These activities create high levels of proximal hamstring tendon compression against the ischial tuberosity which, if not progressed steadily or given enough time for adaptation, can be a primary factor for development of tendinopathy.^{47, 48}

Examination

Physical assessment of an athlete with suspected PHT must first exclude the lumbar spine of any involvement. Lumbar flexion can aggravate the pain of PHT by compressing the proximal hamstring tendon against the ischial tuberosity and reproducing pain.

Neurodynamic testing can also be highly provocative for PHT simply due to the action of stretching the hamstring tendon insertion, but also by compressing the tendon onto the ischial tuberosity. In addition, the close proximity of the proximal hamstring tendon to the sciatic nerve can produce false positives due to biochemical irritation or even in some chronic cases due to fibrous adhesions developing around the sciatic nerve.⁴⁹

PHT pain is usually well reproduced on passive stretch tests such as the bent knee stretch, modified bent knee stretch (Fig. 30.3f) and Puranen-Ovara tests (Fig. 30.3g). These tests have been shown to have high inter-examiner

reliability (range 0.82–0.88) and very high intra-examiner reliability (range 0.87–0.93). Sensitivity and specificity together with predictive values and likelihood ratios of these tests are shown in Table 30.2.⁵⁰

Pain is also well reproduced on strong isometric contraction of the hamstrings, such as in the supine plank position (Fig. 30.3h) and on direct palpation of the ischial tuberosity, which is usually best performed in side-lying to avoid the bulk of the gluteals.⁴⁶

Imaging of the suspected PHT and the ischiogluteal bursa can be used to confirm diagnosis. MRI (Fig. 30.7) and ultrasound scans can often show a thickened tendon, intra-substance heterogeneity and increased signal intensity.⁵¹

Treatment

Research into the conservative treatment of PHT is limited to just a few case studies.^{52, 53} However, using the principles and research into other tendinopathies (Chapter 4) can help guide the management of PHT.

Initial treatment

Treatment for PHT should be focused to reduce pain and minimise compression of the tendon against the ischial tuberosity. Advise the athlete against stretching the hamstrings or gluteal muscles and to avoid positions of sustained hip flexion, such as prolonged sitting, as much as possible.

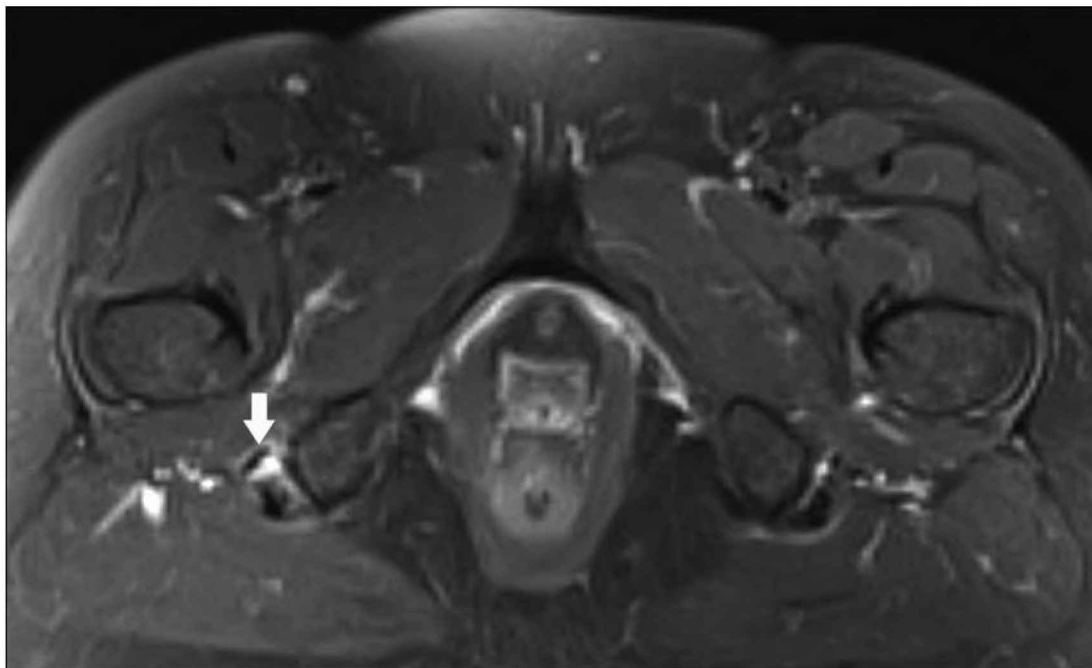


Figure 30.7 T2 fat-suppressed axial MR image showing high T2 signal (arrow) equal to that of fluid involving right common hamstring origin at ischial tuberosity, consistent with partial tear

IMAGING IN SPORTS-SPECIFIC MUSCULOSKELETAL INJURIES, 2016, P.520, ALI GUERMAZI, FRANK W ROEMER, MICHEL D CREMA. © SPRINGER INTERNATIONAL PUBLISHING SWITZERLAND 2016. WITH PERMISSION OF SPRINGER

Treatment can also be directed towards relieving symptoms, using soft tissue techniques on the posterior thigh and gluteals, applying ice and prescribing analgesic medication if needed. Manual therapy techniques, such as deep transverse frictions across the ischial tuberosity and proximal hamstring tendon, are not advocated due to their potential to increase adverse compressive forces. Although they may provide a transient analgesic effect, the forces applied do not break adhesions or increase blood flow significantly.⁵⁴

PRACTICE PEARL

Primary treatment for any tendinopathy must always be staged, progressive loading.⁵⁵

This is no different in PHT and it should begin immediately. Initially in the acute, irritable and painful stages the musculotendinous unit loading should be done well within pain tolerance and in positions that do not compress the tendon.

Isometric contractions can also be very beneficial in the early stages of tendinopathy, producing an analgesic effect as well as maintaining load through the musculotendinous unit. Although the pain reducing effects of isometric contractions are not completely understood, they are believed to be via both peripheral and central nervous system mechanisms.⁵⁶ The most effective 'dosage' of isometric contractions is not fully known, but long duration, 45–60 seconds, at as near as pain tolerance allows to maximal force, seems to have the greatest effect. This should be done regularly and intermittently throughout the day in groups of 3–5 repetitions at a time.⁵⁷ Good examples are the prone hamstring curl (Fig. 30.8) and the supine bridge position (Fig. 30.9a).

Hydrotherapy can also be useful in the early reactive stages to allow loading and movement, while reducing weight-bearing forces and so pain. As pain and irritability

reduces, loading of the hamstrings is progressed steadily, aiming to create the mechanical forces needed to stimulate cellular activity to promote healing as well as restore strength, endurance and capacity to the tendon.⁵⁸

Clear instructions should be given to the athlete that pain during and after tendinopathy loading is normal and to be expected, but it has to be monitored and managed sensibly every 24 hours to ensure daily loading can be completed. Using the visual analogue scale as a guide can be useful. Keeping pain under 4–5/10 when performing rehabilitation exercises seems best to prevent flare ups and allows daily progression.⁵⁹

As the tendon continues to settle, heavier, slower, eccentric resistance training should be used for the hamstring muscles.⁶⁰ Eccentric phases of exercise allow for greater forces over greater durations of time to be tolerated within the musculotendinous unit compared to concentric phases. However, the loading should not focus solely on eccentric contraction. Concentric muscle action is also important, and a mixed and varied program of exercises incorporating both has been shown to achieve good results with tendinopathy.⁶¹

Intermediate stage treatment

Intermediate PHT rehabilitation exercises should incorporate eccentric action as well as increased load and resistance while minimising positions of high tendon compression. Prone hamstring curls (Fig. 30.8) are a simple, effective method to load hamstrings without any adverse compression, and can be easily adapted, depending on symptoms, with either increased or decreased load, intensity, speed of contraction and range of movement.

Body weight exercises, such as bridges, can be simply and easily progressed into more challenging variations. For example, simply doing them in single leg positions (Fig. 30.9a) or elevating the feet (Fig. 30.9b) or varying the amount



Figure 30.8 Prone hamstring curl



(a)
Figure 30.9 (a) Single-leg hamstring bridge



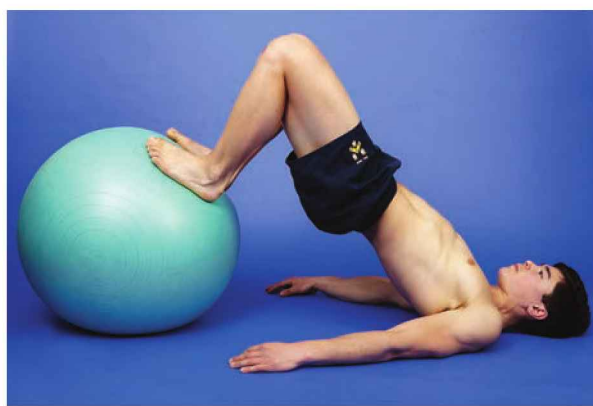
(b)



(d)



(c)



(e)

Figure 30.9 (cont.) (b) Elevated single-leg hamstring bridge (c) to (e) Gym ball hamstring curls

of knee flexion or extension allows for greater or less force in the hamstrings to be felt. Bridging on a gym ball is a good progression for the intermediate stage of rehabilitation. It adds a dynamic component to the exercise as well as involving an element of proprioceptive training (Fig. 30.9c,d,e).

As symptoms continue to reduce, exercises in weight-bearing upright positions can be included. These exercises can also progressively introduce compressive forces by increasing the amount of hip flexion to allow the proximal hamstring tendon to start to adapt to them. Exercises such as 'good mornings' (Fig. 30.10a), deadlifts (Fig. 30.10b) and squats (Fig. 30.10c) are good examples, and can be easily and simply progressed by increasing the amount of hip flexion or increasing external loads carried or performing them on the single leg.

During the tendon loading phase, consideration should be given to identifying and addressing any weaknesses found in the kinetic chain or any biomechanical deficits seen around the lumbopelvic area. Static alignment tests,

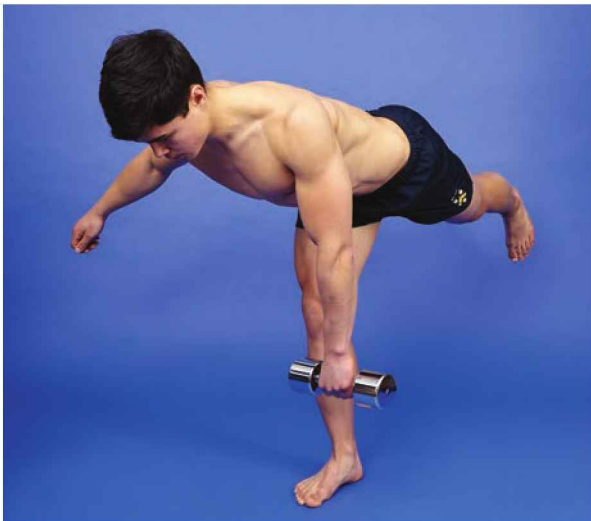
such as planks and their different variations which ask an athlete's lumbopelvic area to resist external forces over a period of time, can give clues to potential weak areas that need addressing.

Dynamic tests, such as single-leg squats and lunges as well as functional tests that mimic an athlete's activity as closely as possible, can give a therapist clues to any possible biomechanical imbalances and lack of proprioceptive ability that may also be a causative factor, while also continuing the appropriate loading for the PHT.

There is no one best or more validated method to assess lumbopelvic strength or biomechanics. Each assessment will be based on, and guided by, the athlete's level of ability and sport, as well as the therapist's own clinical reasoning, experience and preference. Information taken from these tests is to be used with caution and is not to be used in isolation. There is scant evidence and little consensus on the role that biomechanics does or does not have in tendinopathy incidence.

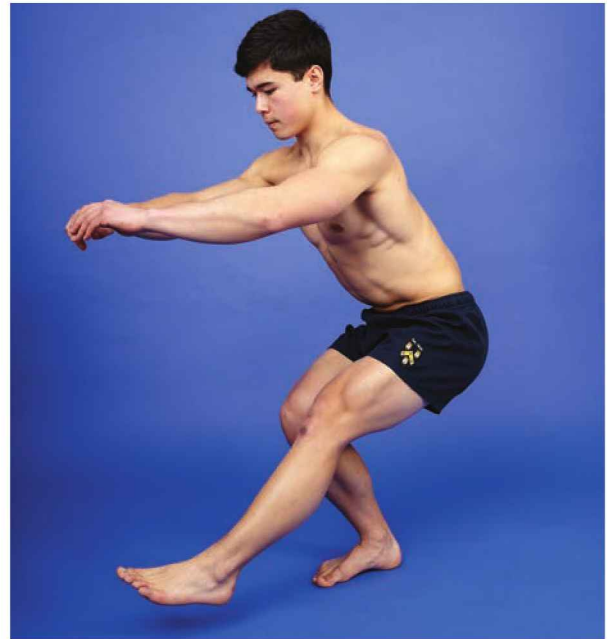


(a)



(b)

Figure 30.10 (a) Loaded double-leg good mornings
(b) Single-leg RDL (Romanian dead lift)/toe touches
(c) Single-leg squat



(c)

sporting activities. They are an essential component for full and safe return to sporting activity. Trying to closely mimic the functional demands, positions, speeds and types of muscle contraction the tendon will experience in its normal sporting activity is key in these final stages of rehabilitation.

When an athlete can return to running tasks depends on the type, frequency and intensity of their training and their pain levels. High-intensity, long-duration activity should be avoided until later, when the symptoms have settled and the musculotendinous unit has progressively adapted. However, less intense, reduced duration running that avoids excessive hip flexion, with instructions to the athlete to avoid hills and use shorter stride lengths to avoid overstriding, allow an athlete to return to running sooner and during their rehabilitation, as long as clear instructions are given to monitor pain levels both during and after activity.

SACROILIAC JOINT DYSFUNCTION

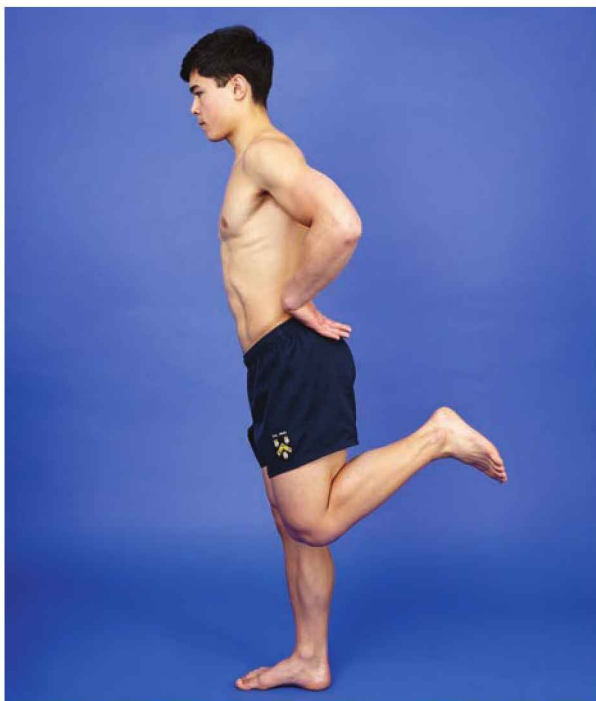
The sacroiliac joint (SIJ) is a highly debated and controversial source of low back and buttock pain in sports, with it often being accused of moving too little, too much or even subluxating or popping out of position.

PRACTICE PEARL

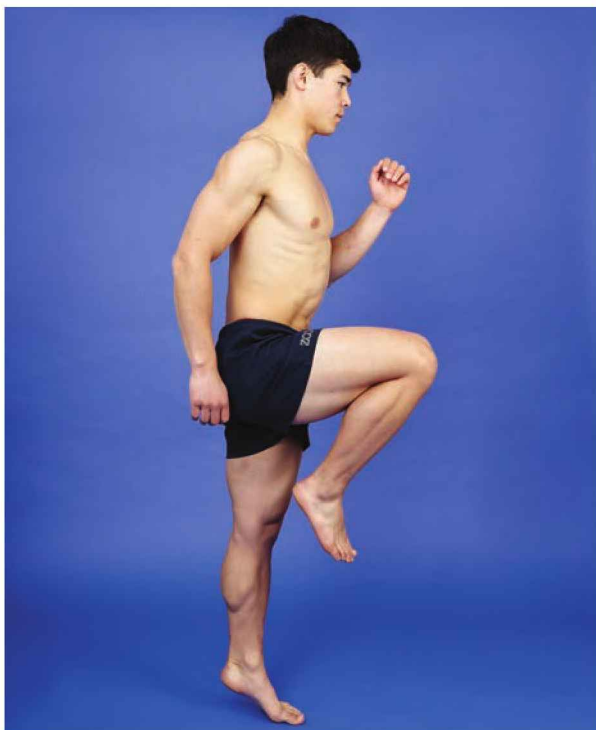
SIJ dysfunction is a broad term and is often used as a scapegoat for low back and buttock pain that has no clear diagnosis.

Final stage of treatment

The final stages of rehabilitation should include progressive higher speed, plyometric and ballistic movements, such as bounding, hopping and jumping, heel flicks and high knee drives (Fig. 30.11). These types of activities expose the musculotendinous unit to sudden changes in eccentric and concentric action, and allow it to adapt to the normal stretch shortening cycle required in most



(a)



(b)

Figure 30.11 (a) Heel flicks (b) High knee drives

The prevalence of SIJ dysfunction varies greatly in the literature, with estimates ranging between 15–30%⁶² and even as high as 50% in certain sporting populations.⁶³

The true prevalence of SIJ dysfunction is extremely difficult to establish, not only due to the complex anatomy of the SIJ with its many ligaments and myofascial structures that can be a source of pain, but also due to the multifactorial nature of low back and buttock pain.

To confuse matters further, many of the gold standard tests, such as imaging and diagnostic local anaesthetic, cannot be relied upon for the SIJ. Imaging alone cannot differentiate symptomatic from non-symptomatic SIJs,⁶⁴ local anaesthetic injections leak out of the joint and affect other tissues and nerve roots around the pelvis.⁶⁵

There is also controversy and disagreement on how and why the SIJ can cause pain. It is commonly believed that poor spinal or pelvic positions or poor posture may be factors in SIJ dysfunction, with excessive anterior or posterior tilt of the pelvis adversely stressing and sensitising either the joint itself, its many ligaments or the myofascial tissue across it.⁶⁶ However, there is no evidence that spinal or pelvic positions or postures have any association with pain or incidence of injury.

Poor strength of the pelvic floor, abdominal or lumbar spine muscles is also often proposed as a causative factor for SIJ dysfunction, in the belief that it allows for excessive SIJ movement and shear forces.^{67, 68}

However, there are doubts around this commonly held theory. There is no correlation between excessive movement of the SIJ and pelvic pain post-partum.⁶⁹ It is even questionable if excessive movement actually occurs at the SIJ, even in those with diagnosed widespread hypermobility joint syndromes and chronic pelvic pain.¹⁸

Finally, despite claims for the need to improve trunk and pelvic strength to increase the stability of the SIJ, we have no idea what amount of trunk and pelvic muscle activity is needed to provide sufficient stability to the SIJ during different tasks, or if it is even a causative factor in developing SIJ pain.

All this uncertainty around the pelvis often leaves the clinician confused and frustrated when trying to determine whether the SIJ is or is not, dysfunctional or a source of pain, let alone how best to manage it.

Functional anatomy

The sacroiliac joints are highly specialised joints that permit a stable, yet slightly flexible platform for the transmission of forces between the lower limb and the rest of the body. The SIJs are large, flat, part synovial, part fibrotic joints that have a highly congruent surface that provides a large friction coefficient, which together with its many strong ligaments, allows for effective and

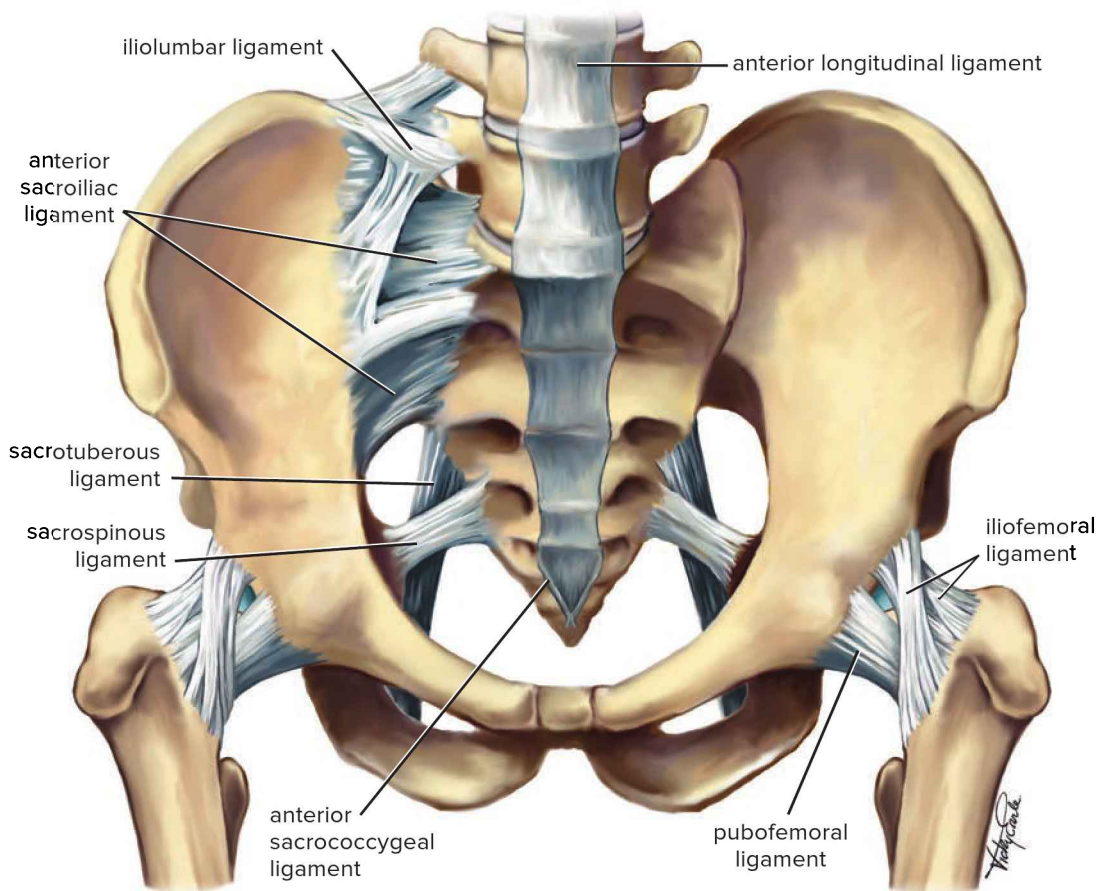


(a)

During movement, SIJ kinematics are highly complex due to the interaction of the many forces from the upper and lower limbs as well as the spine. Rotational movements at the pubic symphysis are an essential component for the pelvis to function as a unit allowing for simultaneous opposite direction movements of the SIJ to occur during gait. During lumbar flexion the sacrum is tilted anteriorly and produces a force on the innominate bones that creates an external rotation and outflaring movement, this is often described as 'nutation'. The opposite occurs during lumbar extension and is usually called 'counter-nutation'.⁷³

Clinical features

An athlete with SIJ dysfunction can report a deep, diffuse, usually unilateral buttock pain that may be triggered after a change in activity level. A full and detailed history of an athlete's training record must be thoroughly explored to locate any potential trigger. The onset, location and sensations of SIJ pain can closely mimic that of a lumbar spine issue. However, SIJ pain is more likely to be reported below the level of L5 and possibly more directly over the SIJ, but rarely over the ischial tuberosity. Pain can also



(b)

Figure 30.12 (cont.) Sacroiliac joint anatomy (b) Anterior view

occasionally be felt bilaterally and sometimes refer distally down the posterior and lateral thigh.⁷⁵

Aggravating factors tend to be activities that increase forces or loads through the SIJ, such as sitting, walking, lumbar spine movements, and sporting tasks. Easing factors can be resting in supine positions, but side-lying can be uncomfortable.

Examination of the athlete with suspected SIJ pain should look globally at the quality of movement of the gait, hip and lumbar spine movements and check to see if any directional preference is produced with repeated movements which may indicate discogenic or lumbar radiculopathy symptoms rather than SIJ dysfunction.⁸ The use of identifying the bony landmarks of the ASIS and PSIS to ascertain if the pelvis is symmetrical or in good alignment has little clinical utility and will not help in the diagnosis or treatment planning.

Other common tests used to assess the SIJ position, alignment or its hypermobility or hypomobility have also been shown to have poor specificity or sensitivity

and again have no place in an evidence-based clinical assessment.^{19, 20, 76} In fact it has been demonstrated that using palpation tests that feel for movement of the SIJ exaggerate the small SIJ movements by up to five times.⁷¹

The best way to determine if the SIJ is a source of buttock pain is with a cluster of pain provocation tests.^{7, 9} These tests can be seen in Figure 30.3 and the sensitivity and specificity of the combinations are shown in Table 30.2.

The active straight leg raise test (ASLR) has also been proposed as a valid test to assess the ability of the SIJ to withstand load (Fig 30.3o).⁹ When it produces a sensation of heaviness or pain, it is thought to highlight trunk or pelvic muscle weakness or altered motor control that reduces SIJ force closure.^{77, 78} However, when a positive ASLR test is measured, the increased movement is found at the symphysis pubis joint not the SIJ.⁷⁹ It is also not known if a positive ASLR test is a cause or an effect of SIJ dysfunction, as experimentally induced SIJ pain in

normal pain-free subjects has been shown to produce a positive test.⁸⁰

Treatment

The management of a patient with suspected SIJ dysfunction must be multimodal to reflect the psychosocial nature of pain, as well as the complex anatomy, biomechanics and relationships that this area has with surrounding structures.

Manual therapy techniques in and around the pelvis, lumbar spine and buttocks, such as soft tissue massage, joint mobilisations and manipulations have been advocated.⁸¹ It must be remembered, however, that the SIJ is highly congruent and well supported by immensely strong ligaments and muscles, therefore these techniques do not and cannot change or alter an SIJ's physical position or affect its biomechanical properties in any way.⁸²

The effects achieved with manual therapy around the SIJ are not fully understood, but it appears they are due more to central and peripheral neuromodulation of the soft tissue and joint mechano- and nociceptors that reduce sensations of stiffness and pain rather than biomechanical or structural mechanisms (see Chapters 6 and 7).⁸³

Other passive treatment modalities such as taping, pelvic belts and braces have also been suggested to offer relief from SIJ pain. They are often thought to help assist weak trunk muscles in the force closure of the pelvis.^{84, 85} However, the exact mechanism of action of SIJ belts is still unknown.⁸⁶

Corticosteroid injections in and around the SIJ can help alleviate pain in the short term, but, as previously mentioned, the injected material often leaks out from the joint and can affect other surrounding tissues, which needs to be considered.

The use of sclerosing prolotherapy injections has been advocated in the management of SIJ pain.⁸⁷ There is a belief that these injections act as a trigger for fibrosis of the sacroiliac ligaments and help restore stability around the joint.⁸⁸ Good quality robust trials using control groups and placebo treatments are lacking with prolotherapy injections, and their use is usually based on anecdotal information.

Although initial treatments for an athlete with SIJ dysfunction can focus on pain relief and symptom modification, ultimately the primary treatment is to restore the capacity of the SIJ and its surrounding tissues to withstand load, stress, shear and strain with a progressive strengthening and functional exercise rehabilitation program.

In the initial stages this may begin with non weight-bearing exercises that focus on the muscles of the posterior chain, the erector spinae, gluteals and hamstrings. Exercises already mentioned in the proximal hamstring tendinopathy section can be used to good effect here.

Exercises such as bridges and their progressions can be combined with exercises that also strengthen the abdominals. Progression of exercises should include moving into weight-bearing positions, focusing on a single plane of movement, such as squats and deadlifts, progressing the resistance and load as symptoms improve. More complex movements involving different and multiple planes of movement, and variable loads and speeds, can be incorporated as the athlete improves, ultimately looking to return the athlete back to full participation in his or her sport.

LESS COMMON CAUSES OF BUTTOCK PAIN

Piriformis syndrome

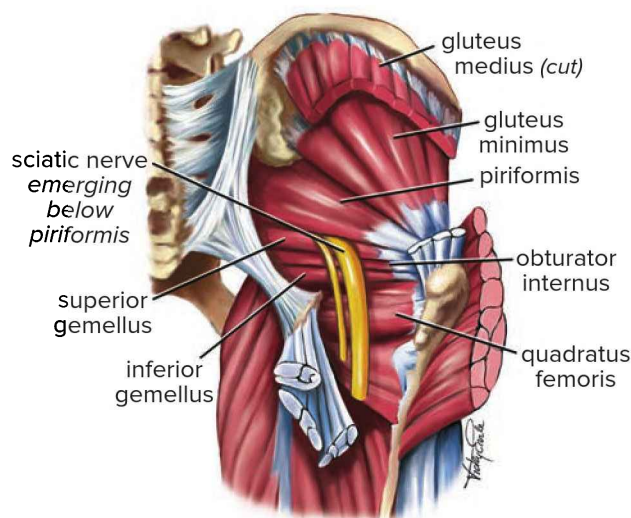
Another highly debated diagnosis of buttock pain is piriformis syndrome or impingement. This diagnosis was first described in 1947 as an enlarged or inflamed piriformis muscle mechanically compressing the sciatic nerve causing buttock and posterior thigh pain.⁸⁹ The anatomy of the piriformis muscle and its relation to the sciatic nerve can be seen in Figure 30.13. Variation in the path of the sciatic nerve around or even through the piriformis has been described as a potential cause for this condition.⁹⁰

It has been proposed that up to 5% of all buttock pain is related to piriformis syndrome.⁹¹ However, there is disagreement over the true prevalence, with some authors arguing that piriformis syndrome is a neurological condition and therefore can only be diagnosed with electro-diagnostic evidence of a neurological deficit.⁹² Others think that piriformis syndrome is a myalgia that occurs secondary to insufficiency or weakness of the larger hip and buttock muscles, such as the gluteals.⁹³

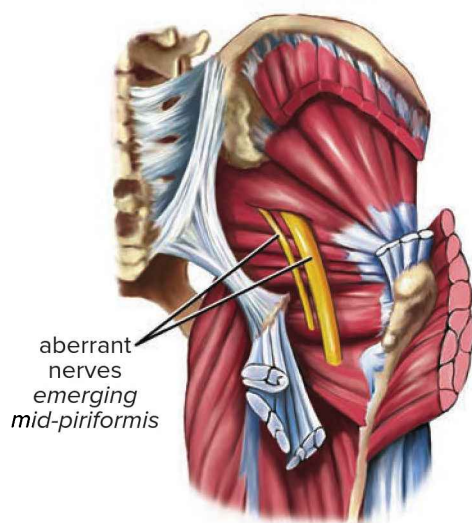
Piriformis syndrome may be more common in sports that involve frequent hip flexion, internal rotation and adduction, such as cycling, skiing, dance and gymnastics. It can occur with a gradual insidious onset as well as through a traumatic event. Pain is usually reported in the mid buttock area and described as deep and diffuse. It tends to increase with seated positions more than standing and pain may be felt when stretching the buttocks.

Clinical examination of an athlete with suspected piriformis syndrome should try to exclude the more common causes of pain such as the lumbar spine, sciatic neuropathy, hamstring tendinopathy, and the SIJ and its ligaments.

The straight leg raise (SLR), the active side-lying piriformis test (Fig. 30.3p) and passive seated piriformis test (Fig. 30.3q) have been investigated to check their sensitivity and specificity to diagnose sciatic nerve



(a)



(b)

Figure 30.13 Course of the sciatic nerve in the buttock
(a) Normal (b) Aberrant

entrapment in the deep gluteal region and can be seen in Table 30.2.⁹⁴

Special tests used to confirm piriformis syndrome are active resistance tests for the piriformis challenging its actions of hip extension from flexion and external rotation, and are often called the PACE and Beatty tests. Other special tests for piriformis syndrome are passive stretch tests of the muscle in positions of hip flexion and internal rotation, often called the FAIR and Freiberg tests. However, these tests have not shown reproducible validity and so must be used with caution as they also stress other structures. These tests can be seen in Figure 30.3.



(a)



(b)

Figure 30.14 Treatment of tight piriformis muscle (a) Soft tissue therapy—piriformis. Sustained longitudinal pressure to the belly of the piriformis muscle, initially in passive external rotation and then moving into internal rotation (b) Stretching of hip external rotators

As with other causes of buttock pain, treatment should first be prioritised to help reduce pain and modify symptoms. This may be achieved with manual therapy techniques, such as soft tissue massage (Fig. 30.14a) and stretching of the piriformis (Fig. 30.14b). Self-applied methods using tennis and golf balls can be taught.⁹⁵ Although the evidence is limited, there can be a role for local anaesthetic injections in the management of piriformis syndrome; however, there seems no advantage in adding corticosteroid to these.⁹⁶

Once symptoms have reduced, the focus should be directed towards strengthening the buttock muscles and their actions of hip extension and abduction. These have already been described in the myofascial pain section in this chapter.

Surgical treatment by open or arthroscopic release has been described in a few cases, but these are uncontrolled case studies and should only be considered as a last resort in chronic, refractory cases.^{97, 98}

Ischiofemoral impingement

The skeletal morphology of the hip joint is well known as a source of impingement in athletes. A rarer type of hip impingement between the lesser trochanter of the femur and the ischium exists, called ischiofemoral impingement (IFI), and this can manifest as atypical groin or buttock pain.

IFI is believed to be due to abnormal or repeated contact between the lesser trochanter of the femur and the ischium, usually when in positions of hip extension, adduction and external rotation, for example when ballet dancers perform their poses and positions.⁹⁹ It could also hypothetically be caused by weakness or fatigue of the hip and pelvic abductors allowing for a Trendelenburg position of excessive pelvic adduction during sports or tasks. This abnormal pelvic and leg position could cause adverse contact, compression and then injury to the quadratus femoris (QF) muscle, a strong adductor and external rotator of the femur and hip stabiliser.

True prevalence of IFI is not fully known, but estimates are around 4% with it predominantly seen in females (84%) and around 40% of cases being bilateral.¹⁰⁰

An athlete with IFI can present with a similar presentation as an adductor or proximal hamstring tendinopathy, with pain felt in the deep buttock and/or groin when they perform sport or activity, that progressively worsens until it is felt in normal daily activity.¹⁰¹ Painful popping and clicking sensations can be reported with IFI and can be misdiagnosed as a snapping iliopsoas or labral injury. IFI can also cause neural symptoms due to the close proximity of the QF muscle to the sciatic nerve.¹⁰²

There are no clinical diagnostic special tests for IFI. IFI should be suspected when movements that compress the QF muscle, such as combined hip extension, adduction and external rotation, reproduce pain; however, these should not be used in isolation and should not be relied upon to diagnose IFI.

To diagnose IFI, an MRI is needed that shows QF muscle oedema and/or a reduced space between the ischial tuberosity and lesser trochanter of the femur (Fig. 30.15).¹⁰³

Treatment

Conservative management options for IFI are initially to provide advice and education regarding activity modification to reduce symptoms and allow for soft tissue recovery. Manual therapy can help reduce soft tissue pain, if needed. Once pain has settled, a progressive strengthening program should be started to address any weakness found around the hip, trunk or lower limb that may have predisposed the QF to adverse compression. Consideration should also be given to biomechanical effects, for example, ensuring there is no adverse or unnecessary crossing of the feet over the mid line when running, as well as to good lumbopelvic control during sporting activities.

The surgical management of IFI is highly controversial as it involves resection of the lesser tuberosity. This has risks to the blood supply to the femoral head as well as risks to the sciatic nerve and so should only be considered in the most extreme cases.¹⁰⁴

Posterior thigh compartment syndrome

Compartment syndrome develops due to increased soft tissue oedema or bleeding that increases tissue pressures within myofascial compartments, compressing

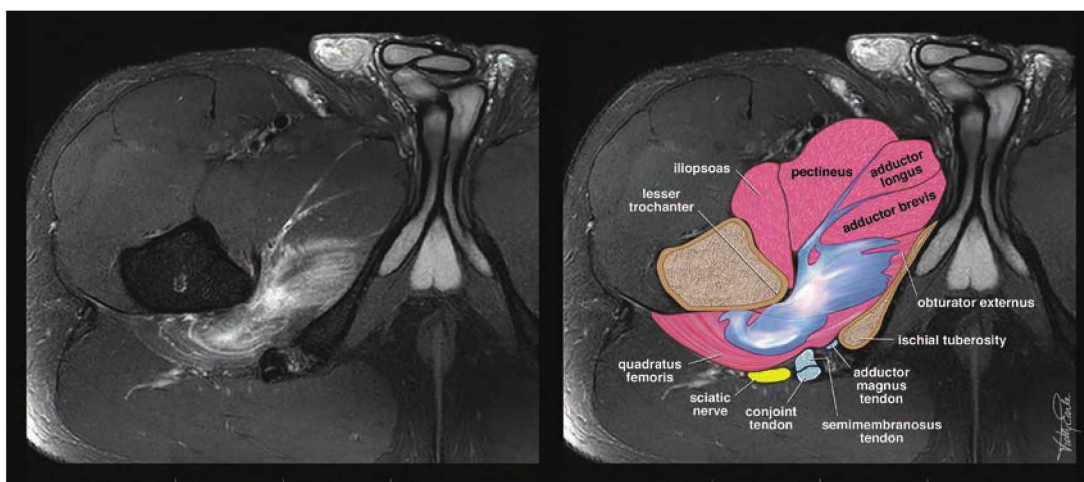


Figure 30.15 MRI showing muscle oedema of the quadratus femoris muscle secondary to ischiofemoral impingement

and compromising the soft tissues, vascular and neural structures.¹⁰⁵ Posterior thigh compartment syndrome is a rare type of compartment syndrome, which causes buttock and posterior thigh pain.

Posterior thigh compartment syndrome can be associated with a traumatic event, such as a blow or a fall onto the buttocks or posterior thigh. It has been reported with acute traumatic hamstring muscle tears or tendon avulsions.^{106, 107} It can also be atraumatic and caused by long periods of immobilisation and inactivity.¹⁰⁸

Posterior thigh compartment syndrome pain is usually diffuse and progressive. It can be accompanied with distal neural and vascular occlusion type symptoms of weakness and paraesthesia. Examination reveals distended, tense and difficult to manipulate soft tissues, possibly with visible bruising if traumatic.

To confirm a diagnosis of compartment syndrome, elevated internal tissue pressures need to be recorded (Chapter 38), with levels between 30–50 mmHg being found to cause muscle necrosis if left more than 4–8 hours.¹⁰⁹ Early diagnosis and careful monitoring of this condition is essential to ensure symptoms do not deteriorate and to avoid any serious complications or adverse effects, such as tissue ischaemia or permanent neural injury.

Conservative management should be directed to help reduce compartment pressures with gentle active and passive movements, and elevation of the limb. Gentle soft tissue massage may help alleviate symptoms. In cases that do not respond with conservative management, or if compartmental pressures worsen, partial surgical fasciotomy is recommended.¹¹⁰

Proximal hamstring tendon rupture

Complete disruption of the proximal hamstring tendon at its attachment onto the ischial tuberosity is rare compared to the more commonly encountered hamstring muscle tears or strains in sport. However, complete proximal hamstring tendon ruptures are seen in sports that have a risk of traumatic sudden hip flexion with the knee remaining in near full extension, such as water skiing, gymnastics and motorcross.¹¹¹

These traumatic mechanisms cause the athlete to experience sudden, severe pain, frequently with a sensation of tearing or popping around the upper posterior thigh and usually the patient will not be able to bear weight.¹¹²

Over the next few hours to days, significant bruising and swelling will emerge extending down the posterior thigh. An observable and palpable deformity may be seen and felt in the posterior thigh on active knee extension when prone. Knee extension will be grossly weak and walking will be difficult due to pain. Confirmation of the extent of the avulsion is best done with MRI (Fig. 30.16) or ultrasound scanning.

Treatment

The decision between conservative and surgical management for proximal hamstring rupture injury is not clear. Some advocate conservative treatment if there is less than 2 cm of retraction seen on imaging.¹¹³ Others suggest all are surgically repaired to ensure complete return of normal muscle strength and power and to maximise chances for return to sport.¹¹⁴

Conservative management of complete hamstring ruptures is often challenging due to the large slow resolving haematoma that can complicate the healing

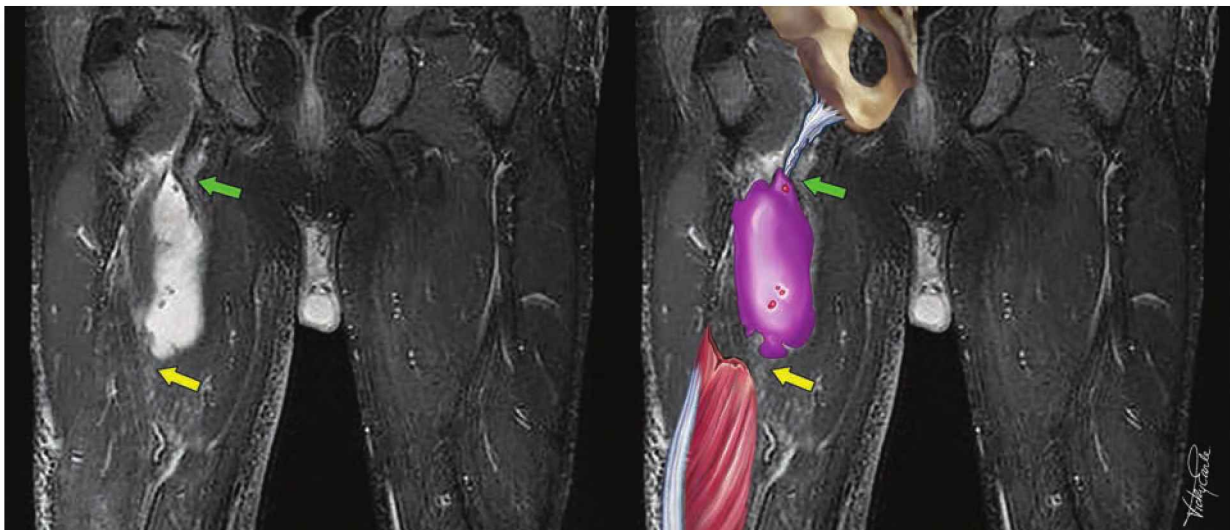


Figure 30.16 MRI showing proximal hamstring rupture. The proximal (green arrow) and distal (yellow arrow) ends of the ruptured tendon are shown with the intervening region of haemorrhage which is the bright signal (white) in the left panel and overdrawn in purple proximal to the retracted muscle in the right panel

process and encourage excessive scarring. This can lead to complications such as adhesions to the sciatic nerve, sometimes called 'hamstring syndrome' and that might even need surgical release.⁴⁹

Conservative treatment for hamstring avulsions should first recognise and ensure that the athlete's expectations are realistic and he or she is fully aware of the extended time this injury will take to recover. Treatment should focus on pain relief and offloading the hamstrings with crutches until weight-bearing can be tolerated. Gentle active and passive movement should be encouraged to prevent joint stiffness and encourage organised scar tissue formation; pool work is a good option here. Manual therapy needs to be used with caution so as not to interfere with normal healing processes. Once weight-bearing is comfortable then a progressive strengthening and conditioning program can begin. However long-term deficits in strength, power and endurance can persist.¹¹⁵

Surgical repair for hamstring avulsions is becoming an increasingly popular management option for both acute and chronic hamstring avulsions, with higher levels of patient satisfaction and return to play, greater strength and endurance, and less pain.¹¹⁶ These injuries are discussed further in Chapter 34.

Avulsion fracture of ischial tuberosity

Avulsion fractures of the ischial tuberosity are similar to complete proximal hamstring tendon ruptures, but are usually seen only in adolescents. They occur when the hamstring tendon fractures the ischial tuberosity, pulling it away from the pelvis.¹¹⁷ They are most commonly seen in adolescent football and gymnastics.¹¹⁸

The age range for these injuries is between 11–17 years old (mean 13.8 years) with the mechanism of presentation usually being exactly the same as a proximal hamstring tendon rupture, a sudden acute pain being felt with a sensation of popping or tearing during a rapid hip flexion movement and an inability to weight-bear.¹¹⁸

Examination is the same as with a proximal hamstring tendon rupture, with bruising and swelling developing over the following few hours and days, weakness and pain on prone knee extension, and possibly a visible and palpable deformity of the hamstrings. Confirmation of an avulsion fracture can be made with an ultrasound or MR scan but it can also be seen on a plain radiograph showing a displaced bony fragment away from the ischium (Fig. 30.17).

Management is usually conservative and should be treated the same as a proximal hamstring tendon rupture, by offloading the hamstrings with crutches and with gentle active and passive movements to avoid joint stiffness. Once able to weight-bear, a progressive strengthening and return to play program can commence. However, if the bony fragment is seen to be displaced more than 2.5 cm,



Figure 30.17 Radiograph of an avulsion fracture of the hamstring tendon from the ischial tuberosity
COURTESY OF KYLE NAGLE

surgery may be recommended to restore congruity of the tendon and ensure return to sport.¹¹⁹

CONDITIONS NOT TO BE MISSED

Spondyloarthropathies

Buttock pain can be one of the first symptoms associated with a group of systemic inflammatory disorders, such as ankylosing spondylitis. There will usually be no history of trauma, rather a non-specific, insidious onset of buttock pain that is resistant to usual conservative treatments lasting for more than 3 months.

Ankylosing spondylitis is more prevalent in Caucasian races and has a mean age of onset around 23 years. It is rarely seen after the age of 40. It has a male-to-female ratio of 3:1. Buttock and spinal pain and stiffness is commonly reported, lasting more than 30 minutes first thing in the morning, and movement and exercise reduces the pain.¹²⁰ These conditions also tend to be associated with multiple joint pains and other enthesopathies in athletes, such as

plantar fasciopathy or tendinopathies, and may have extra articular symptoms as well, usually affecting the eyes.¹²¹

Diagnosis follows observation of a combination of subjective symptoms, imaging and genetic testing for the HLA B27 gene. However, in early manifestations imaging can appear normal. The New York criteria has been suggested as an aid to diagnosis.¹²²

Treatment consists of patient advice and education regarding activity modification, with the emphasis on physical exercise and mobility. Nonsteroidal anti-inflammatory medications, corticosteroid injections and the use of disease-modifying anti-rheumatic drugs (DMARDs) have also been found to help.¹²³

Stress fracture of the sacrum

Sacral stress fractures are a rare and uncommon cause of buttock pain in the athlete, however the diagnosis is often missed or delayed due to a lack of awareness. There is little epidemiological data, but they are most frequently seen in osteopenic female runners, possibly secondary to menstrual and/or eating disorders (Chapter 4).¹²⁴

The clinical presentation can be variable and very similar to a SIJ dysfunction. The main complaint is usually acute low back or pelvic pain, associated with a severe reduction in mobility and a possible radiation to the leg, groin, buttocks and thighs.

Symptoms are increased with weight-bearing activity, and improve with rest and lying supine. Tenderness over the sacral area is common. There can often be associated tenderness over the symphysis pubis, due to the high incidence of pubic rami fractures in association with sacral stress fractures.¹²⁵ Neurological defects are usually absent.

Physical examination may reveal many positive tests for SIJ dysfunction including sacral tenderness on lateral compression; the flexion-abduction-external rotation (FABER) test, Gaenslen's test and the squish test are often positive.¹²⁶ Diagnosis of a sacral or pubic ramus fracture is done with imaging. Radiographs can be inconsistent as they will only show complete fractures and even then abdominal and bowel contents can shadow fracture lines.¹²⁷ MRI (Fig. 30.18) and CT scans are the best options.

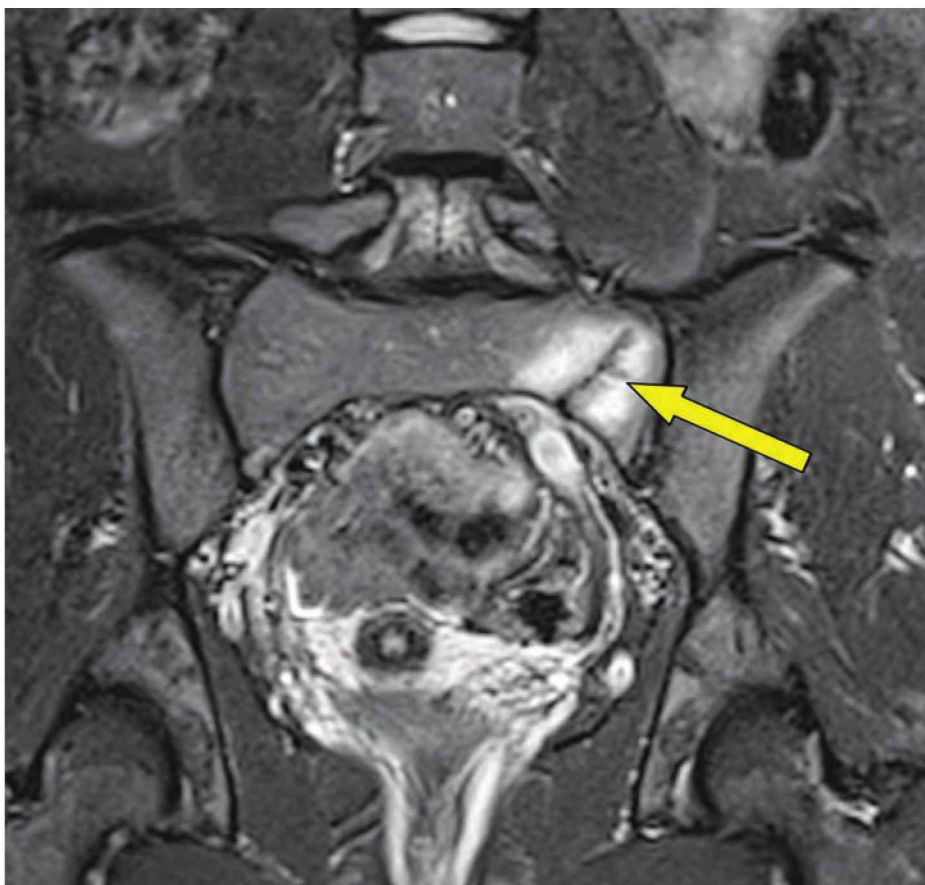


Figure 30.18 Coronal MRI of a sacral stress fracture (arrow)

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There are no guidelines for best practice in the management of sacral stress fractures, with some advocating 3-6 months non weight-bearing, and others just a week or two.¹²⁸

The use of non-steroidal anti-inflammatory medications is not recommended in the presence of fractures due to the potential detrimental effects they can have on bone healing (Chapter 4).^{127, 129}

The principle treatment is a short period of rest to reduce pain and then progressive early mobilisation to

encourage weight-bearing stimulation of osteoblastic activity as well as maintaining bone density, muscle tone and tension, and limiting atrophy and stiffness.

Hydrotherapy is a good early option as body weight can be loaded or offloaded depending on symptoms. As pain continues to settle, a progressive strengthening and conditioning program can be designed and implemented. It has been suggested that return to running activity can be as soon as 2 months after diagnosis, despite pain remaining for around 6 months.¹³⁰

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

